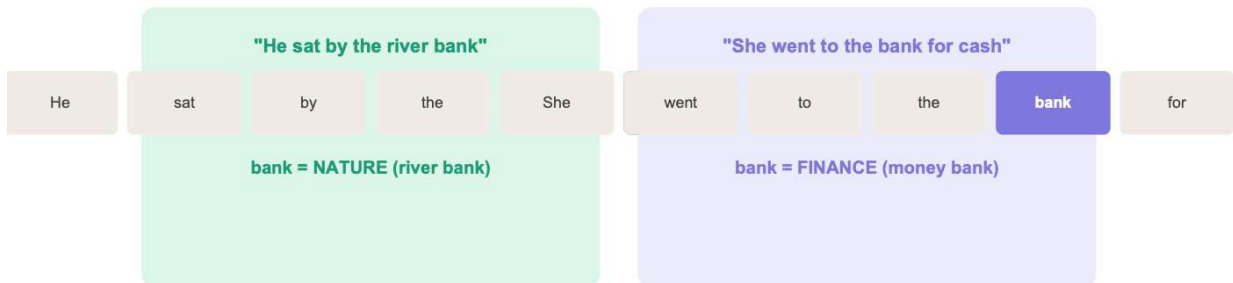


✓ The Problem with Simple Word Vectors

One word, many meanings — static embeddings are stuck with just one vector per word.

In Day 17 we saw how GloVe and Word2Vec give each word a fixed vector. 'King' always has the same 300 numbers. But what about the word 'bank'? It means completely different things depending on its context — and a fixed vector can't capture that. This is called the **polysemy problem**.



What static embeddings miss

GloVe and Word2Vec compute ONE vector for 'bank' by averaging all the contexts it appears in across the training corpus. The result is a blurry average of both meanings — neither river-bank nor money-bank. When your model sees 'river bank' it gets the same vector as 'money bank'. This causes errors in downstream tasks.

The solution: context-aware embeddings Instead of giving each word a fixed vector, we want a model that reads the ENTIRE sentence and produces a DIFFERENT vector for 'bank' depending on whether it appears near 'river' or near 'money'. This is exactly what the Transformer architecture achieves — through a mechanism called Self-Attention, which we'll explore in Section 02.

✓ The Attention Mechanism

Teaching the model which words to focus on when understanding each word. Imagine reading the sentence: 'She picked up the cup and drank from it.' When you reach 'it', your

brain instantly knows 'it = cup'. You didn't re-read every word with equal effort — your attention jumped straight to 'cup'. The Attention Mechanism in neural networks does exactly the same thing, mathematically.

Real-life analogy — attention as a spotlight

Think of a concert. When the guitarist is soloing, your ears (attention) naturally focus on the guitar and tune out the drums. When the singer starts, your attention shifts. The Attention Mechanism gives every word in a sentence its own spotlight that can move around and focus on the most relevant other words.

How Self-Attention works — step by step

For every word in the sentence, Self-Attention asks three questions using three learned vectors — Query (Q), Key (K), and Value (V):

Query (Q) --> What am I looking for? — When processing 'it', Q asks 'what object was mentioned earlier?'

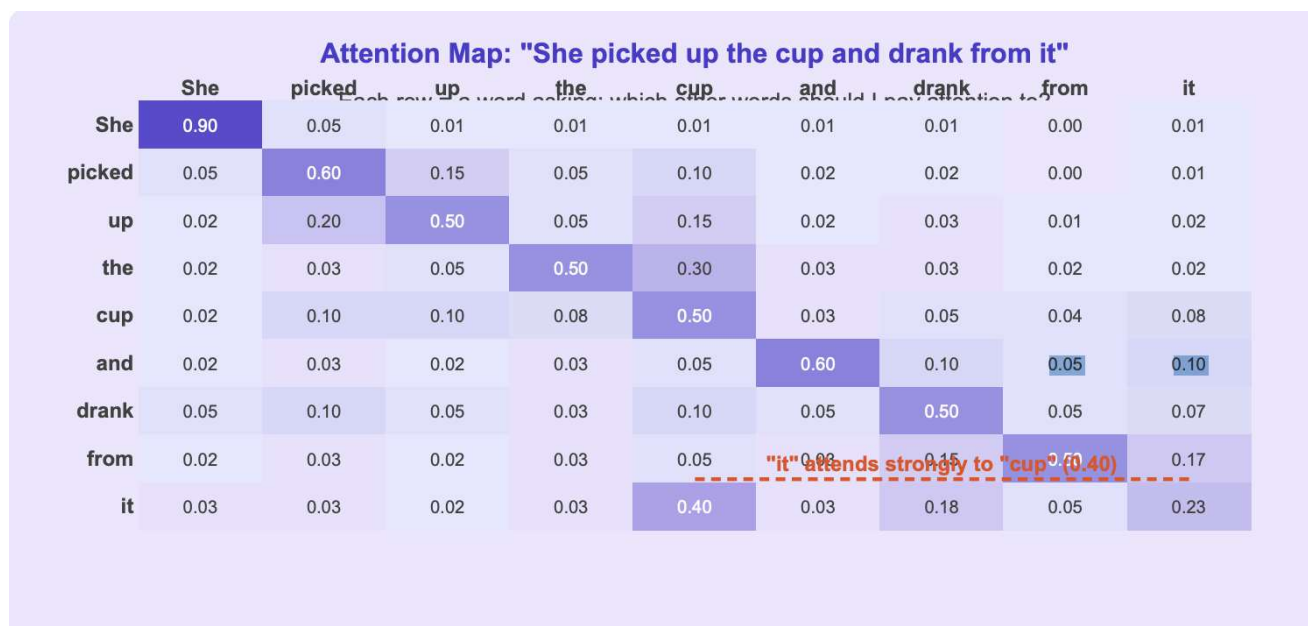
Key (K) --> What do I offer? — 'cup' broadcasts that it's a physical object that can be referred to.

Value (V) --> What information do I actually carry? — 'cup' provides its rich semantic meaning.

The attention score between two words = dot product of Q and K, scaled and softmaxed. High score = strong attention (the word 'it' pays lots of attention to 'cup'). The model then takes a weighted sum of Value vectors.

Visualising attention — colour-coded attention map

Below: darker cells mean the row-word pays more attention to the column-word. Look at the last row ('it') — it has a dark cell under 'cup' (0.40), correctly learning that 'it' refers to 'cup'.



Multi-Head Attention — looking from multiple angles

BERT doesn't just run attention once. It runs it 12 times in parallel, each with different Q/K/V weight matrices. These are called 'heads'. Each head learns to focus on different types of relationships:

Head (example)	What it learns to focus on	Example
Head 1	Subject-verb relationships	'She ... drank'
Head 2	Co-reference resolution	'it' to 'cup'
Head 3	Syntactic dependencies	adjective to noun
Head 4	Entity proximity	'river' near 'bank'
Heads 5-12	Various other linguistic patterns	...

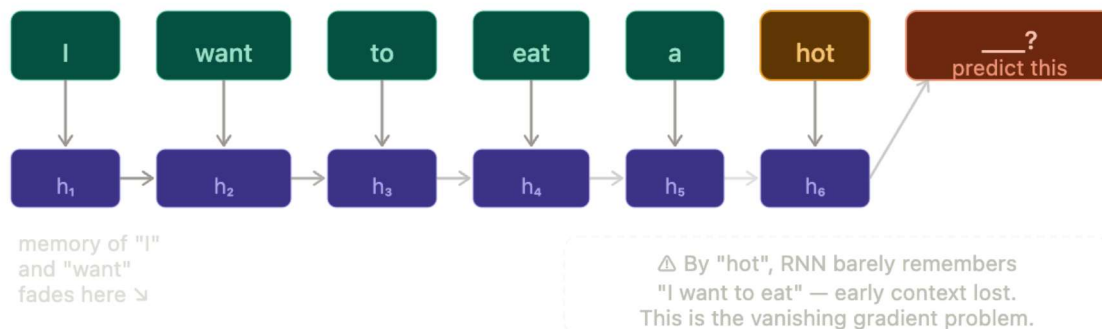
Why attention is better than RNNs Before Transformers, RNNs (Recurrent Neural Networks) processed text word by word, left to right. To connect 'it' to 'cup', the signal had to travel through 4 intermediate words — getting weaker each step (the 'vanishing gradient' problem). Attention connects ANY two words DIRECTLY in one step, regardless of distance. This is why Transformers handle long documents so much better than RNNs.

RNNs (Recurrent Neural Networks) are networks that read sequences one word at a time, left to right, and carry a "memory" from each step to the next. Think of it like reading a sentence while only keeping notes in a tiny notebook — by the time you reach the end, early details can be forgotten.

Real-life example: Predicting the next word as you type When you type "I want to eat a hot ___", an RNN reads each word one by one, passing its memory forward. But by the time it reaches "hot", the early context ("I want to eat") may have faded — this is called the vanishing gradient problem.

RNN — reads left to right, one word at a time

Sentence: "I want to eat a hot ____"



BERT — reads ALL words at once
strong: "eat" + "hot" → predicts "food" or "dog"

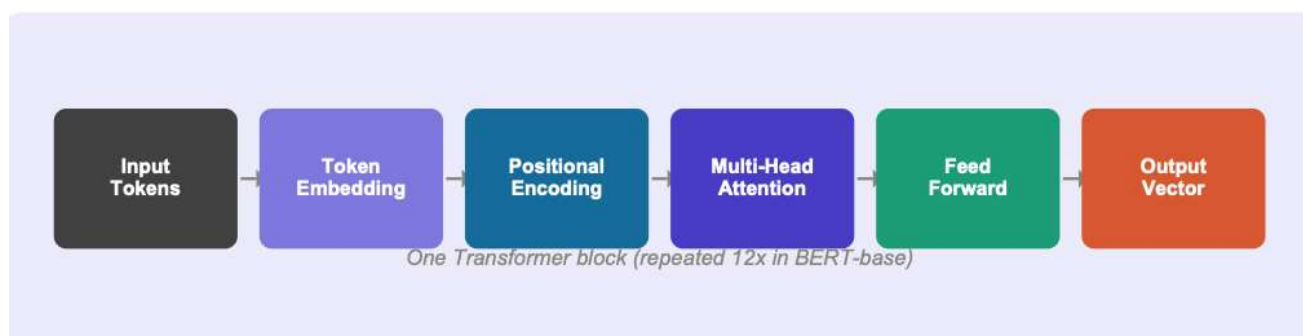
Same sentence: "I want to eat a hot ____"



RNN	BERT
Reads left → right, one word	Reads all words at once
Memory fades over distance	Attention connects any two words
Slower, older approach	Faster, modern (Transformer-based)

Transformers — The Architecture that Changed NLP

A **Transformer** is made up of multiple layers of **Multi-Head Attention** and **Feed-Forward Networks**. Each layer helps words better understand their relationship with other words in the sentence. As the data passes through the layers, the word representations become more context-aware. The model gradually learns deeper meanings and connections between words. In **BERT-base**, there are **12 such layers**. After all 12 layers, the model has a rich understanding of each word in the context of the entire sentence.



Key components

Token Embedding --> Each word (token) is converted to a 768-D vector. Like Word2Vec but learned jointly with the model.

Positional Encoding --> 'She loves him' is not 'He loves her'. Since Attention has no built-in notion of order, we add a sinusoidal position signal to each token embedding to encode word order.

Multi-Head Attention --> 12 attention heads run in parallel. Each learns different linguistic relationships. Outputs are concatenated and linearly projected back to 768-D.

Feed-Forward Network A simple 2-layer fully connected network applied to each position. Adds non-linearity and extra capacity. Output is still 768-D.

Layer Norm + Residuals --> Each sub-layer has a residual (skip) connection and layer normalisation, ensuring stable training even with 12+ stacked layers.

BERT-base vs BERT-large vs modern models

Model	Layers	Heads	Parameters	Context window
BERT-base	12	12	110 M	512 tokens
BERT-large	24	16	340 M	512 tokens
GPT-3	96	96	175 B	4,096 tokens
Claude Sonnet 4.6	—	—	~100 B+	200,000 tokens

✓ BERT — A Pre-trained Language Brain

Bidirectional Encoder Representations from Transformers — Google AI, 2018. BERT took the Transformer encoder and pre-trained it on a massive scale: all of English Wikipedia (2.5 billion words) plus BookCorpus (800 million words). The result is a model that deeply understands English grammar, facts, relationships, and nuance — before you show it a single labelled example. You then fine-tune this pre-trained brain on your specific task.

What makes BERT 'Bidirectional'?

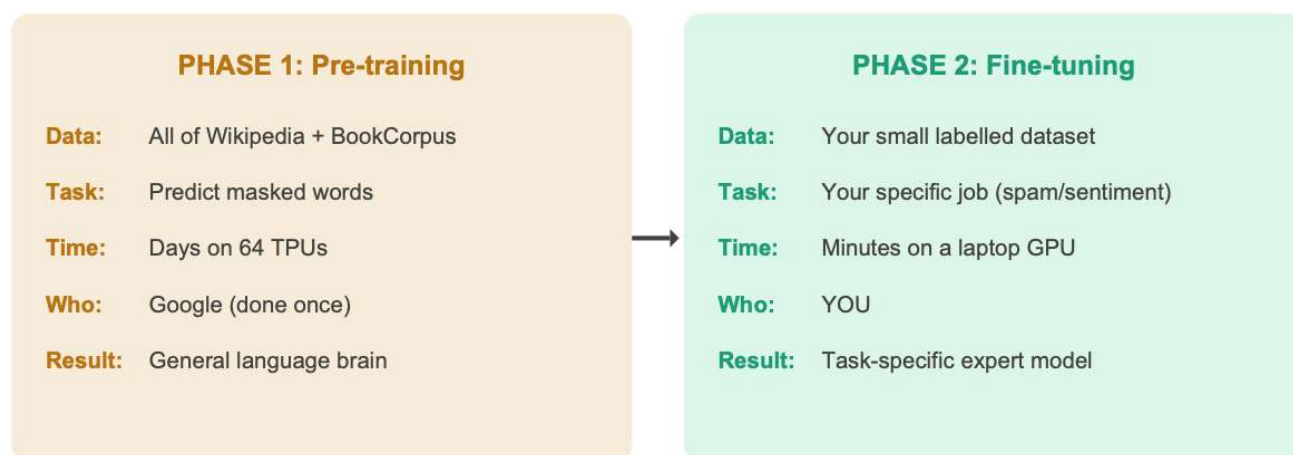
Previous models like GPT-1 read text left-to-right only. BERT reads the ENTIRE sentence simultaneously — left and right context at every position. When understanding the word 'bank' in position 5, BERT can see all words before AND after it. This bidirectionality is why BERT dramatically outperformed earlier models on tasks like question-answering.

How BERT was pre-trained — two clever tasks

Task 1: Masked Language Model (MLM) ---> 15% of input tokens are randomly replaced with [MASK]. BERT must predict the original word. Example: 'The [MASK] sat on the mat' predicts 'cat'. This forces BERT to understand full bidirectional context.

Task 2: Next Sentence Prediction (NSP) ---> Given two sentences A and B, predict if B actually follows A in the original text. This teaches BERT about inter-sentence relationships — vital for QA and NLI tasks.

Pre-training vs Fine-tuning — the two-phase approach



What can you fine-tune BERT for?

Question Answering	Find start/end of answer span	SQuAD benchmark
Text Classification	[CLS] token multi-class head	Spam / Sports / Politics
Sentence Similarity	Compare [CLS] embeddings	Duplicate question detection

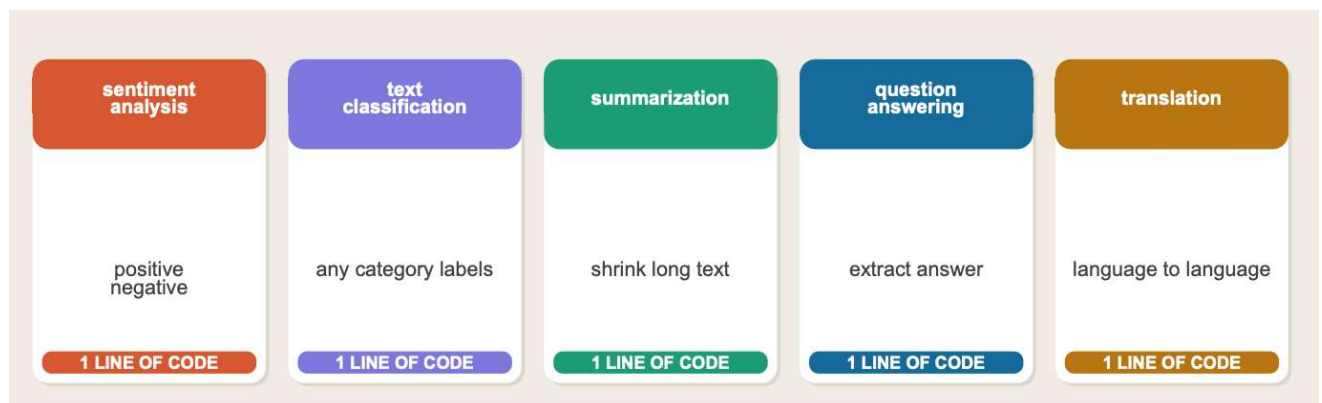
Question Answering	Find start/end of answer span	SQuAD benchmark
Text Classification	[CLS] token multi-class head	Spam / Sports / Politics
Sentence Similarity	Compare [CLS] embeddings	Duplicate question detection

Real-life analogy — BERT as a medical graduate

A fresh medical graduate has spent 5 years studying all of human anatomy and medicine (pre-training on Wikipedia). You then hire them and give them 2 weeks of specific training on dermatology (fine-tuning on your dataset). They become an excellent dermatologist far faster than training someone from scratch — because their general knowledge (the BERT base) is already world-class. That's exactly what transfer learning with BERT does for NLP.

✓ HuggingFace Pipelines — One Line of Code

HuggingFace is the 'GitHub for AI models' — a platform with 500,000+ pre-trained models freely available. Their `pipeline()` function wraps any model with automatic tokenisation, inference, and post-processing. What used to take 200 lines of code now takes 3.



```
# Install HuggingFace Transformers
!pip install transformers torch
from transformers import pipeline
# The pipeline() function handles everything:
# 1. Downloads the model weights (~400 MB, cached after first run)
# 2. Loads the tokeniser (converts text to token IDs)
# 3. Runs inference through the Transformer model
# 4. Post-processes and returns human-readable results
```

[Show hidden output](#)

Pipeline 1 — Sentiment Analysis Classifies text as POSITIVE or NEGATIVE with a confidence score. Uses DistilBERT fine-tuned on the SST-2 movie reviews dataset by default.

```
from transformers import pipeline
senti = pipeline('sentiment-analysis')
reviews = [
    'This movie was absolutely breathtaking! A masterpiece.',
    'Terrible acting, boring plot. Wasted 2 hours of my life.',
    'It was okay. Nothing special but not bad either.',
    'Shah Rukh Khan was brilliant. Loved every minute!',
    'Ram is more attractive than jayanth ani evaro chepparu',
    'KSRM is a great college in asia',
    'charan is bad '
```

```

]
results = senti(reviews) # pass a list – processes all at once
for review, result in zip(reviews, results):
    label = result['label'] # 'POSITIVE' or 'NEGATIVE'
    score = result['score'] # confidence: 0.0 to 1.0
    print(f'{label} ({score:.1%}) | {review[:50]}...')
# POSITIVE (99.8%) | This movie was absolutely breathtaking! A masterp...
# NEGATIVE (99.9%) | Terrible acting, boring plot. Wasted 2 hours of m...
# NEGATIVE (55.2%) | It was okay. Nothing special but not bad either...
# POSITIVE (99.1%) | Shah Rukh Khan was brilliant. Loved every minute!.
```

[transformers] No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 7
Using a pipeline without specifying a model name and revision in production is not recommended.

Loading weights: 100% 104/104 [00:00<00:00, 9.08it/s]

```

POSITIVE (100.0%) | This movie was absolutely breathtaking! A masterpi...
NEGATIVE (100.0%) | Terrible acting, boring plot. Wasted 2 hours of my...
POSITIVE (99.6%) | It was okay. Nothing special but not bad either....
POSITIVE (100.0%) | Shah Rukh Khan was brilliant. Loved every minute!...
POSITIVE (100.0%) | Ram is more attractive than jayanth ani evvaro chep...
POSITIVE (100.0%) | KSRM is a great college in asia...
NEGATIVE (100.0%) | charan is bad ...
```

Pipeline 2 — Text Classification (Zero-shot) Zero-shot classification lets you define your OWN category labels — the model was never trained on your specific categories but still classifies correctly by understanding the semantic meaning of your labels.

```

from transformers import pipeline
# Zero-shot: classify without any task-specific training
classifier = pipeline('zero-shot-classification',
model='facebook/bart-large-mnli')
text = 'ksrm is a college of engineering . in aiml we have best students in non crt but cse are worst except few(nen itey a
labels = ['technology','course','computer science','artificial intellignece']
result = classifier(text, candidate_labels=labels)
for label, score in zip(result['labels'], result['scores']):
    print(f'{label:15} {score:.1%}')
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:112: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to

Loading weights: 100% 515/515 [00:00<00:00, 5169.73it/s]

```

technology      57.7%
course          28.7%
artificial intellignece 10.7%
computer science 2.9%
```

Pipeline 3 — Summarisation

```

import transformers
print(transformers.__file__)
print(transformers.__version__)
```

Show hidden output

```
!pip install --force-reinstall transformers==4.44.2 tokenizers==0.19.1 --break-system-packages
```

Show hidden output

```

from transformers import pipeline

summariser = pipeline(
    task="summarization",
    model="facebook/bart-large-cnn"
)
```

```

long_text = """
The Indian Space Research Organisation (ISRO) successfully launched its
latest Earth observation satellite from the Satish Dhawan Space Centre in
Sriharikota. The satellite carries high-resolution cameras capable of
imaging at 0.5-metre resolution, making it among the most advanced Earth
observation tools deployed. The data will be used for agriculture
monitoring, disaster management, and urban planning.
"""
```



```
summary = summariser(
    long_text,
    max_length=20,
    min_length=15,
    do_sample=False,
    truncation=True
)

print(summary[0]["summary_text"])
```

The cache for model files in Transformers v4.22.0 has been updated. Migrating your old cache. This is a one-time only operation.
0/0 [00:00<?, ?it/s]

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as the secret `HF\_TOKEN` in your Colab secrets, and restart this notebook.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

warnings.warn(
model.safetensors: 100% 1.63G/1.63G [00:15<00:00, 144MB/s]
generation_config.json: 100% 363/363 [00:00<00:00, 31.5kB/s]
vocab.json: 899k/? [00:00<00:00, 15.1MB/s]
merges.txt: 456k/? [00:00<00:00, 25.4MB/s]
tokenizer.json: 1.36M/? [00:00<00:00, 30.0MB/s]
/usr/local/lib/python3.12/dist-packages/transformers/tokenization_utils_base.py:1601: FutureWarning: `clean\_up\_tokenization`
warnings.warn(
Asking to truncate to max_length but no maximum length is provided and the model has no predefined maximum length. Default to the Indian Space Research Organisation (ISRO) successfully launched its latest Earth observation satellite.

Pipeline 4 — Question Answering Extractive QA: given a context passage and a question, the model finds and returns the exact span of text that answers the question.

```
from transformers import pipeline
qa = pipeline('question-answering',
model='deepset/bert-base-cased-squad2')
context = '''
Sachin Tendulkar played for India for 24 years from 1989 to 2013.
He scored 100 international centuries and 15,921 Test runs.
He was awarded the Bharat Ratna in 2014.
'''

questions = [
'How many international centuries did Sachin score?',
'When did Sachin Tendulkar retire?',
'What award did he receive in 2014?',
]
for q in questions:
    answer = qa(question=q, context=context)
    print(f'Q: {q}')
    print(f'A: {answer["answer"]} (confidence: {answer["score"]:.1%})')
# Q: How many international centuries did Sachin score?
# A: 100 (confidence: 94.2%)
```

config.json: 100% 508/508 [00:00<00:00, 23.3kB/s]
model.safetensors: 100% 433M/433M [00:05<00:00, 106MB/s]
Some weights of the model checkpoint at deepset/bert-base-cased-squad2 were not used when initializing BertForQuestionAnswering
- This IS expected if you are initializing BertForQuestionAnswering from the checkpoint of a model trained on another task or with another architecture
- This IS NOT expected if you are initializing BertForQuestionAnswering from the checkpoint of a model that you expect to be initialized with those weights

tokenizer_config.json: 100% 152/152 [00:00<00:00, 11.6kB/s]
vocab.txt: 213k/? [00:00<00:00, 10.9MB/s]
special_tokens_map.json: 100% 112/112 [00:00<00:00, 8.97kB/s]
Q: How many international centuries did Sachin score?
A: 100 (confidence: 95.2%)
Q: When did Sachin Tendulkar retire?
A: 2013 (confidence: 67.0%)
Q: What award did he receive in 2014?
A: Bharat Ratna (confidence: 97.4%)

Pipeline 5 — Translation

```
from transformers import pipeline
# Helsinki-NLP models for 1000+ language pairs
# Model naming: Helsinki-NLP/opus-mt-{source}-{target}
```

```

en_hi = pipeline('translation',
model='Helsinki-NLP/opus-mt-en-te')
texts = [
'Machine learning is changing the world.',
'I love cricket and Bollywood movies.',
]
for text in texts:
    translated = en_hi(text)[0]['translation_text']
    print(f'EN: {text}')
    print(f'HI: {translated}')
# EN: Machine learning is changing the world.
# HI: mshyn lrnng duniya bdl rhi hai.
# For French: pipeline('translation', model='Helsinki-NLP/opus-mt-en-fr')

```

Show hidden output

English → French

```
pipeline('translation', model='Helsinki-NLP/opus-mt-en-fr')
```

English → German

```
pipeline('translation', model='Helsinki-NLP/opus-mt-en-de')
```

English → Telugu

```
pipeline('translation', model='Helsinki-NLP/opus-mt-en-te')
```

✓ Going Further — Fine-tuning BERT (Saturday-Sunday Task)

When the default pipeline isn't enough, train BERT on your own labelled data. The `pipeline()` function uses models already fine-tuned on generic datasets. For domain-specific tasks — medical text, legal documents, Hinglish slang, product reviews in a niche category — you'll get better results by fine-tuning BERT on your own data. Here's a minimal working example:

Fine-tuning with HuggingFace Trainer API — Part 1: Setup

```

from transformers import (AutoTokenizer,
AutoModelForSequenceClassification,
TrainingArguments, Trainer)
from datasets import Dataset
# --- Step 1: Prepare your labelled dataset ---
texts = ['Great film!', 'Terrible movie.', 'Average, nothing special.',
'Best film of the year!', 'Waste of time and money.']
labels = [1, 0, 0, 1, 0] # 1 = positive, 0 = negative
# --- Step 2: Load tokeniser ---
# 'bert-base-uncased' = lowercase BERT, 110M parameters
tokenizer = AutoTokenizer.from_pretrained('bert-base-uncased')
def tokenise(batch):
# padding=True: pad shorter sequences to the longest in the batch
# truncation=True: cut sequences longer than max_length
    return tokenizer(batch['text'], padding=True,
truncation=True, max_length=128)
dataset = Dataset.from_dict({'text': texts, 'label': labels})
dataset = dataset.map(tokenise, batched=True)
# --- Step 3: Load pre-trained BERT + classification head ---
# num_labels=2 adds a 2-class head on top of BERT's [CLS] output
model = AutoModelForSequenceClassification.from_pretrained(
'bert-base-uncased', num_labels=2
)

```

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 4.24kB/s]

config.json: 100% 570/570 [00:00<00:00, 52.1kB/s]

vocab.txt: 232k/? [00:00<00:00, 5.25MB/s]

tokenizer.json: 466k/? [00:00<00:00, 12.6MB/s]

Map: 100% 5/5 [00:00<00:00, 62.15 examples/s]

model.safetensors: 100% 440M/440M [00:07<00:00, 75.8MB/s]

Some weights of BertForSequenceClassification were not initialized from the model checkpoint at bert-base-uncased and are new. You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

```

# --- Step 4: Define training configuration ---
args = TrainingArguments(
output_dir = './results',

```



```

num_train_epochs = 3, # 3 passes through training data
per_device_train_batch_size = 4, # process 4 examples at a time
learning_rate = 2e-5, # small LR - avoid forgetting pre-training
logging_steps = 10,
save_strategy = 'no',
)
# --- Step 5: Train ---
trainer = Trainer(
    model = model,
    args = args,
    train_dataset = dataset,
)
trainer.train() # fine-tunes BERT on your labelled data
# --- Step 6: Save and reload as a pipeline ---
model.save_pretrained('./my_sentiment_model')
tokenizer.save_pretrained('./my_sentiment_model')
# Use your custom fine-tuned model exactly like any other pipeline
from transformers import pipeline
my_pipeline = pipeline('sentiment-analysis', model='./my_sentiment_model')
print(my_pipeline('This Bollywood film was a pure delight!'))
# [{'label': 'LABEL_1', 'score': 0.97}] -> LABEL_1 = positive

```

wandb: WARNING The `run\_name` is currently set to the same value as `TrainingArguments.output\_dir`. If this was not intended

/usr/local/lib/python3.12/dist-packages/notebook/notebookapp.py:191: SyntaxWarning: invalid escape sequence '\/'

```

|_| |' _ \ / _ \ | _/ -_)

```

wandb: (1) Create a W&B account

wandb: (2) Use an existing W&B account

wandb: (3) Don't visualize my results

wandb: Enter your choice: 1

wandb: You chose 'Create a W&B account'

wandb: Create an account here: <https://wandb.ai/authorize?signup=true&ref=models>

wandb: After creating your account, create a new API key and store it securely.

wandb: Paste your API key and hit enter:

wandb: ERROR Invalid API key: API key must have 40+ characters, has 1.

wandb: (1) Create a W&B account

wandb: (2) Use an existing W&B account

wandb: (3) Don't visualize my results

wandb: Enter your choice: 3

wandb: You chose "Don't visualize my results"

wandb: Using W&B in offline mode.

wandb: W&B API key is configured. Use `wandb login --relogin` to force relogin

Tracking run with wandb version 0.27.1

W&B syncing is set to `offline` in this directory. Run `wandb online` or set WANDB_MODE=online to enable cloud syncing.

Run data is saved locally in /content/wandb/offline-run-20260612_014304-6jppsbb3

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true

super().__init__(loader)

 [6/6 00:07, Epoch 3/3]

Step Training Loss

[{'label': 'LABEL_1', 'score': 0.6326547861099243}]